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PLA_2 and PI3K/PTEN Pathways Act in Parallel to Mediate Chemotaxis

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SUMMARY

Directed cell migration involves signaling events that lead to local accumulation of PI(3,4,5)P_3, but additional pathways act in parallel. A genetic screen in *Dictyostelium discoideum* to identify redundant pathways revealed a gene with homology to patatin-like phospholipase A_2. Loss of this gene did not alter PI(3,4,5)P_3 regulation, but chemotaxis became sensitive to reductions in PI3K activity. Likewise, cells deficient in PI3K activity were more sensitive to inhibition of PLA_2 activity. Deletion of the PLA_2 homolog and two PI3Ks caused a strong defect in chemotaxis and a reduction in receptor-mediated actin polymerization. In wild-type cells, chemoattractants stimulated a rapid burst in an arachidonic acid derivative. This response was absent in cells lacking the PLA_2 homolog, and exogenous arachidonic acid reduced their dependence on PI3K signaling. We propose that PLA_2 and PI3K signaling act in concert to mediate chemotaxis, and metabolites of PLA_2 may be important mediators of the response.

INTRODUCTION

Chemotaxis, the directed movement of cells along extracellular gradients, is an intriguing and critical cell biological response. A variety of chemoattractants direct cells to proper locations in developing embryos and guide the connections between cells of the nervous system. After embryogenesis, directed cell movements mediate morphogenetic processes such as remodeling of the vascular system and wound healing. Moreover, leukocytes traffic between the vascular and lymphatic systems and migrate from the blood toward sites of infection. In addition to these roles in normal physiology, inappropriate cell migration is the basis for pathological conditions including cancer metastasis and chronic inflammatory diseases such as atherosclerosis, asthma, and arthritis (Hill et al., 2000; Heldin and Westermark, 1999; Baggiolini, 2001; Park et al., 2002; Rao et al., 2002).

The chemotactic response encompasses cell motility, polarity, and directional sensing, all of which are events that involve local phosphatidylinositol-3,4,5-trisphosphate (PI[3,4,5]P_3) production and degradation at the plasma membrane (Parent and Devreotes, 1999; Rickert et al., 2000; Chung et al., 2001; Wang et al., 2002; Heit et al., 2002). The levels of PI(3,4,5)P_3 are regulated by PI 3-kinases (PI3Ks), the PI 3-phosphatase PTEN (Buczynski et al., 1997; Hirsch et al., 2000; Li et al., 2000; Sasaki et al., 2000; Funamoto et al., 2001, 2002; Iijima and Devreotes, 2002; Stocker et al., 2002), and PI 5-phosphatases, or SHIPs. When *Dictyostelium discoideum* cells are exposed to a gradient of chemoattractant, PI3Ks and PTEN bind to the membrane at the front and rear, respectively, PI(3,4,5)P_3 selectively accumulates at the leading edge, and new F-actin-filled pseudopodia are extended at corresponding sites. Unable to degrade PI(3,4,5)P_3, *pten*$^-$ cells fail to sharply localize the phosphoinositide and extend projections in multiple directions at once, which impairs polarity and chemotaxis (Iijima and Devreotes, 2002; Iijima et al., 2004). These observations have established a link between local plasma membrane PI(3,4,5)P_3 levels, actin polymerization, and pseudopodia extension. This role for local PI(3,4,5)P_3 in polarity and directional sensing appears to be conserved throughout phylogeny in cells such as *Drosophila melanogaster* hemocytes, human neutrophils and fibroblasts, neurons, and a variety of embryonic cells (Stramer et al., 2005; Wang et al., 2002; Wu et al., 2000; Schneider and Haugh, 2005; Chadborn et al., 2006; Montero et al., 2003; Wymann et al., 2000).

In spite of these observations, an essential requirement for local PI(3,4,5)P_3 accumulation has been surprisingly difficult to establish. In *Danio rerio*, PI(3,4,5)P_3 is uniformly elevated around the entire perimeter of chemotaxing primary germ cells (Dumstrei et al., 2004). Abrogation of PI3K activation has no effect on migration of *D. melanogaster* border cells. Migration of *PI3K*$\gamma^{-/-}$ mouse macrophages in response to chemokines is only partially impaired, with inhibition ranging from 50% to 90%, depending on the chemokine examined (Hannigan et al., 2002). T cells and B cells have been reported to be sensitive to PI3K inhibitors (Reif et al., 2004), but under more physiological conditions, T cells are reported to be relatively insensitive to PI3K inhibitors (Sotsios et al., 1999;

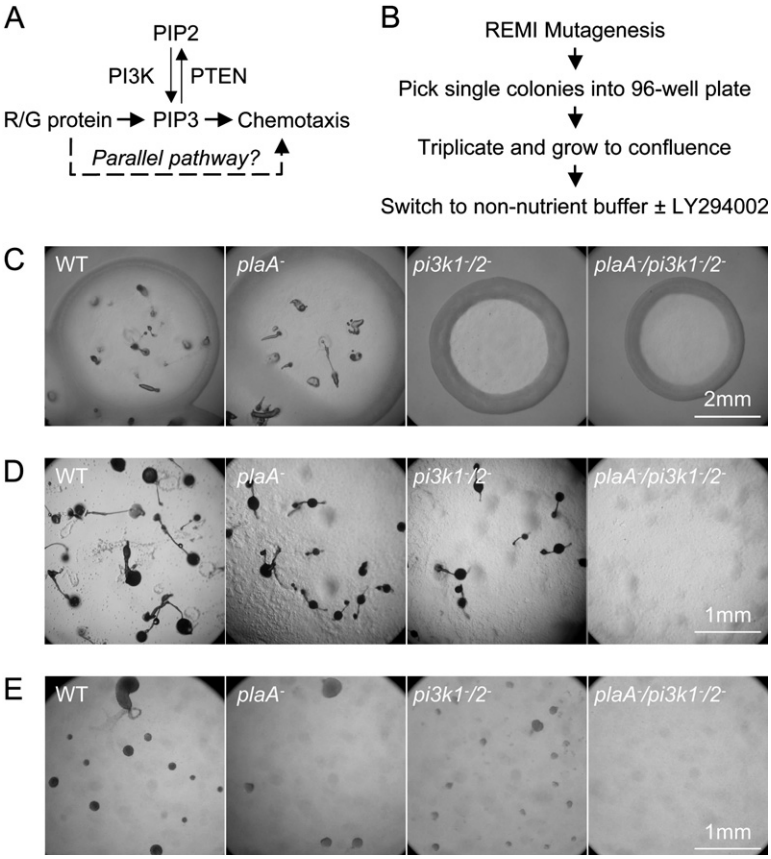

Figure 1. Isolation of Mutant Affecting Parallel Pathway

(A) A model for regulation of chemotaxis. The dashed line shows a putative pathway that acts in parallel with the PI3K/PTEN pathway.

(B) Scheme of the genetic screen. After mutagenesis, individual phenotypically wild-type colonies were separately plated in a 96-well format, replicated, and induced to differentiate in non-nutrient buffer containing different concentrations of LY. Mutants with altered sensitivity to LY are scored based on their ability to aggregate.

(C–E) Phenotypes of wild-type, *plaA*$^-$, *pi3k1*$^-$/*2*$^-$, and *plaA*$^-$/*pi3k1*$^-$/*2*$^-$ cells on (C) bacteria lawn, (D) non-nutrient agar, and in (E) the under-buffer assay.

Smit et al., 2003; Cinamon et al., 2001; Ward, 2004). In *D. discoideum*, while overproduction of PI(3,4,5)P_3 disrupts chemotaxis, loss of PI3K activity (deletion of two out of five PI3Ks), leading to a more than 95% reduction of PI(3,4,5)P_3, hardly affects the ability of cells to sense and orient in chemotactic gradients (Buczynski et al., 1997; Funamoto et al., 2001; Huang et al., 2003; Chen et al., 2003; Loovers et al., 2006). Taken together, these studies in different systems suggest that, although local PI(3,4,5)P_3 plays an important role in chemotaxis, there are likely redundant mediators.

Our studies of chemoattractant-mediated actin polymerization, one of the primary biochemical responses underlying pseudopod extension, have also pointed to alternative pathways. We examined the biphasic responses of PI(3,4,5)P_3 accumulation and actin polymerization during perturbations that decreased or elevated PI(3,4,5)P_3 levels in *D. discoideum*. In cells with deficient PI3K activity, the second phase of actin polymerization was blocked, but the first phase was unaffected. Similar results were obtained with EGF stimulation of breast carcinoma cells (Mouneimne et al., 2004). Taken together, these observations suggest that the second phase of actin polymerization is mediated by PI(3,4,5)P_3, but that the first phase is regulated differently (Chen et al., 2003; unpublished data).

Based on these findings, we have speculated that there are additional pathways, not merely additional PI3Ks, downstream of G protein activation, that act in parallel with the PI3K/PTEN system, controlling actin polymerization, pseudopod extension, and directional movement (Figure 1A). Parallel pathways could explain why elevation of PI(3,4,5)P_3 causes a dramatic defect while its removal does not. Lowered PI(3,4,5)P_3 would not be expected to cause a significant phenotype unless the parallel pathways are also inhibited. Accordingly, we designed a screen to search for mutants in which chemotaxis is selectively impaired when PI3K is inhibited. This procedure uncovered a gene with homology to a patatin-like phospholipase A_2 (PLA_2), suggesting that this enzyme acts in parallel with PI3K pathways to mediate chemotaxis.

RESULTS

Isolation of Mutants Defective in Aggregation in the Presence of PI3K Inhibitors

We screened for components in pathways that act in parallel with PI3K/PTEN, as outlined in Figures 1A and 1B. Restriction enzyme-mediated insertional mutagenesis (REMI) was used to generate random insertions in a population of wild-type cells (Adachi et al., 1994). Mutagenized cells were clonally plated onto bacteria lawns, and cells from phenotypically wild-type single colonies were transferred into 96-well plates. These were grown to confluency, triplicated, then switched to non-nutrition buffer

containing no, low (30–50 μM), or high (>150 μM) concentrations of the PI3K inhibitor LY294002 (LY). During starvation, untreated cells begin to sense and secrete cAMP, which directs chemotactic migration into large, tight aggregates containing several million cells. The low concentration of inhibitor does not significantly alter this process, while the high concentration blocks it, causing the cells to remain as a monolayer, or to form small, loose aggregates. Phenotypes were visually inspected under a dissection microscope after 24 hr. Of 7043 insertional mutants, we found 10 mutants that were unable to aggregate in the presence of the low concentration of inhibitor and 6 mutants that were able to resist the high concentration. The wild-type cells did not show an aggregation defect until 80 μM LY and completely failed at 120 μM. We were able to identify the insertion site on two sensitive and one resistant mutant. Sensitive clones, S1 and S2, formed small aggregates at 20–40 μM LY and failed completely at 80 μM, while resistant clone R1 aggregated normally up to 160 μM. Consistent behavior was observed when each of the clones was retested with another PI3K inhibitor, wortmannin. In this report, we focus on characterization of mutant S1.

A Patatin-like Phospholipase A_2 Homolog Is Required for Aggregation in the Presence of LY

We recovered the tagged sequence integrated into the genome of S1, and we determined the position of the insertion site. It was found to be 191 bp upstream of the start codon of a predicted ORF that shared sequence homology with the group VI PLA_2 family of proteins (Figure S1A; see the Supplemental Data available with this article online). These enzymes are found both in plants, where they are known as patatin or patatin-related PLA_2 (Holk et al., 2002), and in animals, where they are referred to as Ca^{2+}-independent PLA_2 ($iPLA_2$, group VI PLA_2). The family includes enzymes that use triacylglycerol as well as phospholipids as substrates. We suggest the names *plaA* and PLA_2A for the gene (DDB0205513) and its product, respectively. We found 13 additional predicted patatin-like or group VI PLA_2 homologs in the *D. discoideum* genome (Figure S1B). Based on sequence analyses, there are no secreted or cytosolic PLA_2s in this organism (Schaloske and Dennis, 2006). By the use of sequence comparison, the 14 PLA_2s can be roughly divided into 3 groups (Figure S1C). At least two additional ORFs (DDB0219540 and DDB0231758) are closely related to *plaA*. Some of the ORFs have additional domains, including pleckstrin-like (DDB0184540), protein kinase-like (DDB0185559), and growth factor motifs (DDB0185561).

To confirm that the LY-sensitive phenotype was caused by inactivation of *plaA*, we targeted it by homologous recombination, thereby replacing part of the gene with a blasticidin-resistance marker (Bsr) (Figure S2A). The disruption of *plaA* recapitulated the phenotype of the original insertional mutant. On bacteria lawns, individual *plaA*$^-$ cells proliferated to form wild-type-sized plaques that differentiated normally to form fruiting bodies (Figure 1C). Similarly, the *plaA*$^-$ cells were able to aggregate normally

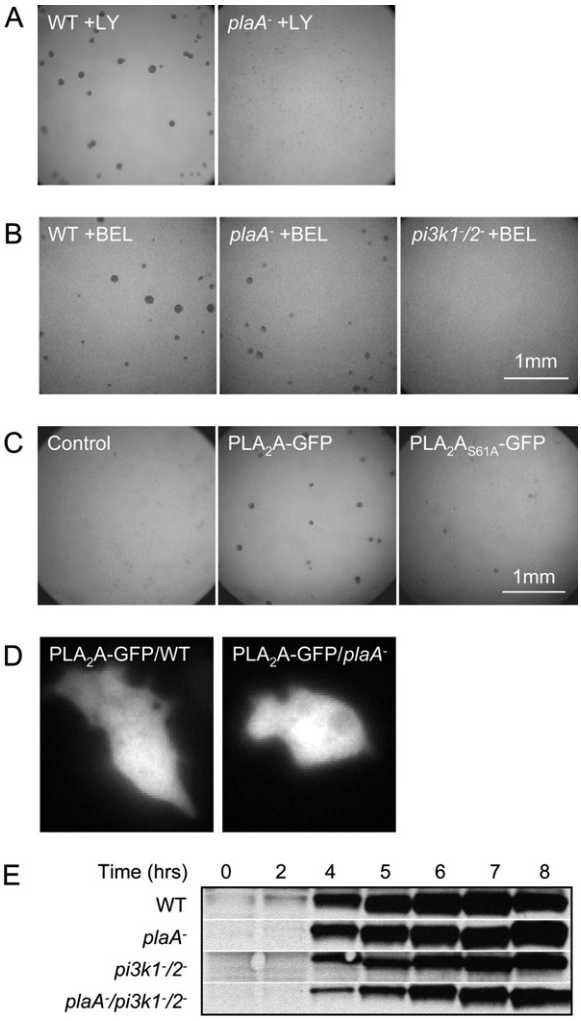

Figure 2. PLA_2A and PI3K Act in Parallel to Mediate Aggregation

(A and B) An under-buffer assay of indicated cells in the absence or presence of 60 μM LY or 5 μM BEL is shown.

(C) Complementation of *plaA*$^-$ cells by GFP-tagged wild-type and the inactive (S61A) version of PLA_2A protein. The phenotype of cells in the under-buffer assay in the presence of 20 μM LY is shown.

(D) Fluorescence images of wild-type and *plaA*$^-$ cells expressing PLA_2A-GFP.

(E) The immunoblot shows expression of the cAMP receptor cAR1 in wild-type, *plaA*$^-$, *pi3k1*$^-$/2$^-$, and *plaA*$^-$/*pi3k1*$^-$/2$^-$ cells during development. Bands are 40 kD. All experiments were repeated at least three times.

and form fruiting bodies like wild-type cells after 24 hr on non-nutrient agar (Figure 1D). However, in the presence of low LY in the under-buffer aggregation assay, the *plaA*$^-$ cells displayed an aggregation defect while wild-type cells were not affected (Figure 2A). Therefore, the phenotypes of the *plaA*$^-$ cells were similar to those of the original mutant, proving that a deficiency in *plaA* causes the LY-sensitive phenotype.

When treated for 5 min with 5 μM bromoenol lactone (BEL), an $iPLA_2$ inhibitor (Ackermann et al., 1995), the

pi3k1$^-$/2$^-$ cells failed to aggregate, while both treated wild-type and *plaA*$^-$ cells still aggregated normally in under-buffer assays (Figure 2B). Thus, cells lacking *plaA* require PI3K activity for aggregation, and, conversely, those deficient in PI3K1 and PI3K2 require PLA_2 activity.

PLA_2A and PI3K Act in Parallel to Mediate Aggregation

If PLA_2A and PI3K are in redundant pathways as suggested, a combined deficiency would be expected to display a severe defect compared to either single disruption. We disrupted *plaA* in cells lacking two of the five "class I-like" PI3Ks (*pi3k1*$^-$/2$^-$ cells) (Buczynski et al., 1997; Funamoto et al., 2001) by using a hygromycin-resistance marker (data not shown; Egelhoff et al., 1989). As predicted, the *plaA*$^-$/*pi3k1*$^-$/2$^-$ cells were unable to aggregate on non-nutrient agar, while wild-type, *plaA*$^-$, and *pi3k1*$^-$/2$^-$ cells developed normally and formed fruiting bodies (Figure 1D). Upon closer examination, we noticed that the *plaA*$^-$/*pi3k1*$^-$/2$^-$ cells formed loose aggregates on non-nutrient agar after 24 hr, suggesting they were motile. Similar results were observed in the under-buffer assays, in which wild-type and *plaA*$^-$ cells aggregated normally, *pi3k1*$^-$/2$^-$ cells formed small aggregates, and *plaA*$^-$/*pi3k1*$^-$/2$^-$ cells remained as a monolayer (Figure 1E). Thus, whereas neither PLA_2A nor PI3K1/PI3K2 is essential, each becomes indispensable in the absence of the other.

To further explore the interdependence of PLA_2A and PI3K signaling, we made an expression construct of PLA_2A tagged with GFP at the C terminus (PLA_2A-GFP). When expressed in *plaA*$^-$ cells, this construct was able to rescue the LY-sensitive phenotype in the under-buffer aggregation assay (Figure 2C). At 20 μM LY, aggregation of *plaA*$^-$ cells was inhibited, but when PLA_2A-GFP was expressed in the *plaA*$^-$ cells, they aggregated normally. The rescue was partial since PLA_2A-GFP/*plaA*$^-$ cells failed to aggregate at 60 μM LY, a concentration that did not block the aggregation of wild-type cells (data not shown). We also expressed an inactive version of PLA_2A-GFP (S61A) in which the catalytic site was mutated. Based on fluorescence and immunoblot of GFP, S61A was expressed at the same level as wild-type PLA_2A-GFP, but it completely failed to rescue the LY-sensitive phenotype in the presence of 20 μM LY. When PLA_2A-GFP was expressed in *plaA*$^-$/*pi3k1*$^-$/2$^-$ cells, it restored the capacity of cells to aggregate in the under-buffer assay (data not shown). PLA_2A-GFP appeared to localize largely in the cytosol in both wild-type and *plaA*$^-$ cells, and its distribution did not change upon cAMP stimulation (Figure 2D).

The phenotypes observed in the *plaA*$^-$ and *plaA*$^-$/*pi3k1*$^-$/2$^-$ cells are not due to defects in development. Since the *plaA*$^-$ cells formed fruiting bodies at the same time after starvation as wild-type cells both on non-nutrient agar and on bacteria lawns, the developmental program is not altered (Figures 1C and 1D). Similar sensitivity to LY was observed regardless of whether it was added at the start or after 5 hr of starvation. As a further indicator of development, the time course of expression of the chemoattractant receptor cAR1 was examined by immunoblot (Figure 2E). cAR1 levels start to increase at 3–4 hr after starvation and reach a peak at 5–6 hr, when chemotactic phenotypes are typically assessed. Wild-type and *plaA*$^-$ cells had essentially identical patterns of expression of cAR1. Although the *pi3k1*$^-$/2$^-$ and *plaA*$^-$/*pi3k1*$^-$/2$^-$ cells expressed slightly lower amounts of cAR1, the time courses were similar to those of wild-type. (An interesting phenomenon was observed in the course of the experiments to determine cAR1 levels. Initially, *pla*$_2$*a*$^-$/*pi3k1*$^-$/2$^-$ cells displayed the wild-type time course and level of cAR1 during development. However, after culture for 4 weeks, cAR1 expression levels gradually declined to the levels seen here, while those in *pla*$_2$*a*$^-$ and *pi3k1*$^-$/2$^-$ cells remained similar to those in wild-type. A similar phenomenon had been observed in *gα2*$^-$ and *gβ*$^-$ cells [unpublished data].) Therefore, the nonaggregation phenotype of the *plaA*$^-$/*pi3k1*$^-$/2$^-$ cells was more likely due to a defect in chemotaxis or cell-cell signaling than to a failure to differentiate.

PLA_2A Is Not Required for $PI(3,4,5)P_3$ Production

To monitor the dynamics of $PI(3,4,5)P_3$ production, PH_{crac}-GFP was expressed in wild-type, *plaA*$^-$, *pi3k1*$^-$/2$^-$, and *plaA*$^-$/*pi3k1*$^-$/2$^-$ cells (Figure 3; Movies S1–S6). PH_{crac}-GFP binds to $PI(3,4,5)P_3$ and translocates to the plasma membrane upon cAMP stimulation (Iijima and Devreotes, 2002; Huang et al., 2003; Lemmon, 2003). In response to a uniform cAMP stimulus, PH_{crac}-GFP in *plaA*$^-$ cells displayed a biphasic response, as previously reported for wild-type cells (Chen et al., 2003; Loovers, et al., 2006). PH_{crac}-GFP was recruited to the entire plasma membrane within seconds and returned to the cytosol after 30 s; then, a second phase, localized to membrane extensions, ensued for the next several minutes (Figure 3A and data not shown). When cells were pretreated with 30 μM LY and stimulated with cAMP, the first phase of $PI(3,4,5)P_3$ accumulation was still detectable, while the second phase was nearly absent, indicating a similar extent of inhibition in each case. In the *pi3k1*$^-$/2$^-$ and *plaA*$^-$/*pi3k1*$^-$/2$^-$ cells, PH_{crac}-GFP did not translocate to the membrane upon cAMP stimulation. Quantitative results were obtained when the translocation of PH_{crac}-GFP was assessed by pelleting of membranes from cell lysates (Figures 3B and 3C). The extent of inhibition with 20 and 100 μM LY was essentially identical in the wild-type and *plaA*$^-$ cells. Thus, neither the dynamics of $PI(3,4,5)P_3$ production nor its sensitivity to inhibition was altered by the loss of PLA_2A, indicating that the enzyme does not regulate the PI3K/PTEN pathway.

PLA_2A Is Required for Chemotaxis in the Absence of PI3K Activity

To directly observe the chemotactic responses, cells were exposed to chemoattractant gradients created by a cAMP-filled micropipette, and directed migration was observed by phase-contrast microscopy. Both wild-type and *plaA*$^-$ cells were examined in the presence and absence of 30 μM LY (Figure 4A; Movies S7–S10). During

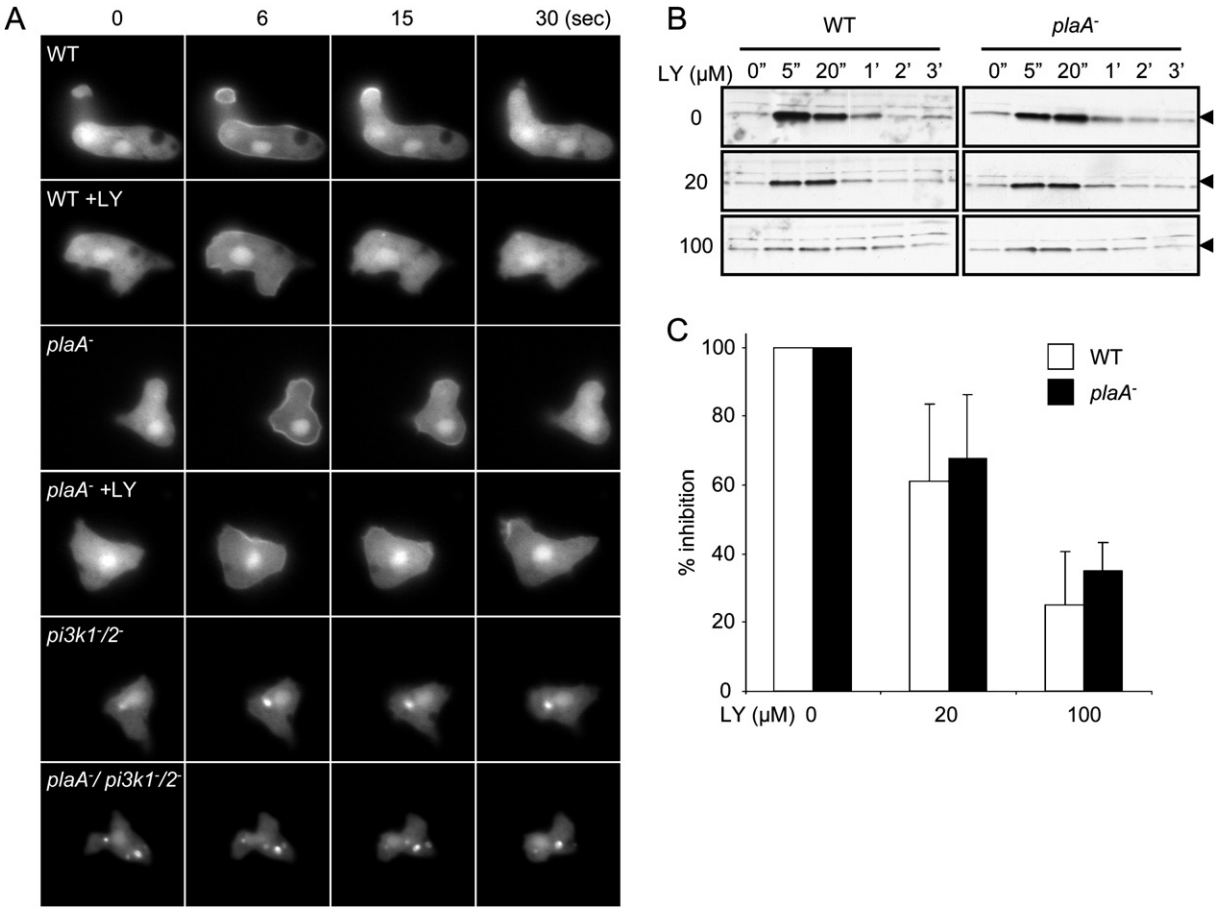

Figure 3. PI(3,4,5)P_3 Accumulation in Wild-Type, *plaA*$^-$, *pi3k1*$^-$/*2*$^-$, and *plaA*$^-$/*pi3k1*$^-$/*2*$^-$ Cells

(A) Chemoattractant-induced translocation of PH_{crac}-GFP to the plasma membrane is observed by epifluorescence microscopy. Images were captured at the indicated time points after uniform stimulation of 1 μM cAMP. When indicated, 30 μM LY was added 10 min before stimulation.

(B) Immunoblotting shows the amounts of membrane-associated PH_{crac}-GFP (arrow) at the indicated time points after addition of 1 μM cAMP.

(C) Quantification of membrane-bound PH_{crac}-GFP was accomplished by densitometry and NIH image. The integrated amount of membrane-bound PH_{crac}-GFP above the prestimulated level was determined. Normalized values relative to controls in the absence of LY are shown. The experiment was repeated three times. Error bars indicate means and standard deviations.

10 min of observation, wild-type cells polarized in the cAMP gradient and moved toward the tip of the micropipette, forming streams by attaching to each other during the directional movement. When cells were treated with LY, they initially lost polarity, then recovered within a few minutes, extended pseudopods, and started to move toward the source of chemoattractant. The LY-treated cells did not form streams during the course of the experiment. The *plaA*$^-$ cells were able to polarize and carry out chemotaxis, similar to wild-type cells. However, addition of 30 μM LY to the *plaA*$^-$ cells severely inhibited chemotaxis and movement. In the presence of LY, only a fraction of *plaA*$^-$ cells were motile, and even fewer moved toward the micropipette. These observations suggest that the migration of the *plaA*$^-$ cells is sensitive to inhibition of PI3K activity.

We next examined the chemotaxis of the *plaA*$^-$/*pi3k1*$^-$/*2*$^-$ cells (Figure 4B; Movies S11–S14). As we observed previously, wild-type cells and *plaA*$^-$ cells were well polarized, formed streams, and moved toward cAMP. Although the *pi3k1*$^-$/*2*$^-$ cells were less polarized, they did line up and migrate toward the micropipette efficiently. Furthermore, treatment of *pi3k1*$^-$/*2*$^-$ cells with LY did not significantly impair their chemotactic response (Movie S15). In contrast, the *plaA*$^-$/*pi3k1*$^-$/*2*$^-$ cells showed strong defects in maintaining cell shape and in directed migration. While the polarities of these cells were similar to those of the *pi3k1*$^-$/*2*$^-$ cells, many of them ignored the direction of the gradient, moved randomly, and changed direction frequently. A few cells very close to the micropipette moved in the correct direction, although sometimes they subsequently turned away. For each experimental condition, we quantified the speed and chemotactic index from three sets of independent experiments (Figure 4C). The mean chemotactic indices showed differences. In wild-type cells, the mean was close to 1, indicating that the cells moved straight toward the needle. In *plaA*$^-$ and *pi3k1*$^-$/*2*$^-$ cells, the indices were

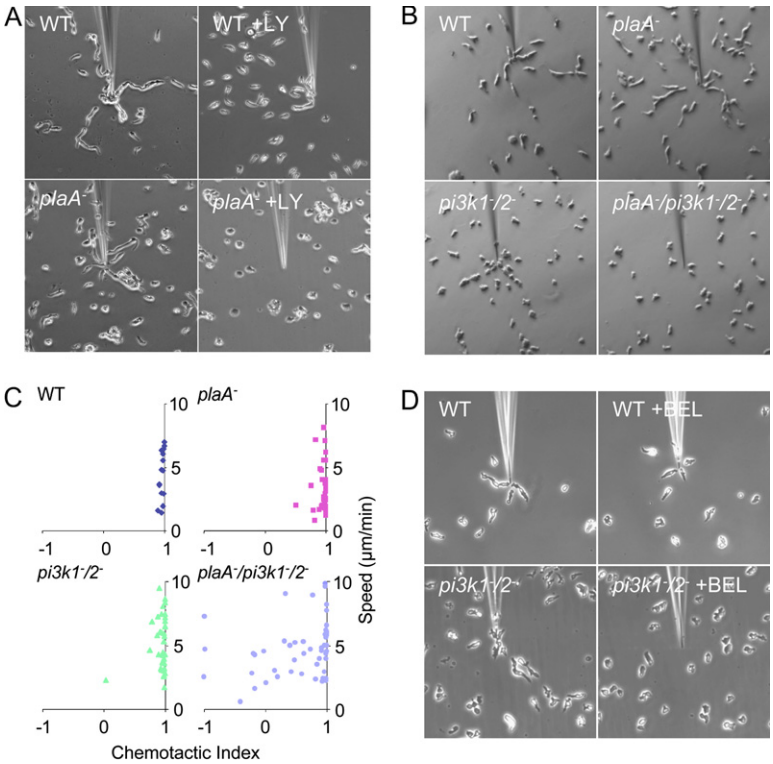

Figure 4. Chemotactic Responses toward a Micropipette Filled with 10 μM cAMP

(A–D) Cells were observed at 10 s intervals for 10 min. There was a random distribution of cells at the onset; images at 10 min are shown. (A) Chemotaxis of wild-type and *plaA*$^-$ cells was observed in the absence or presence of 30 μM LY by using phase-contrast microscopy. LY was added as indicated 10 min before the start of the experiments. (B) Chemotactic responses of wild-type, *plaA*$^-$, *pi3k1*$^-$/*2*$^-$, and *plaA*$^-$/*pi3k1*$^-$/*2*$^-$ cells were observed by DIC microscopy. (C) The speed (S), calculated from the distance a cell covered divided by the experimental time, and the chemotactic index (CI), defined as the cosine of the angle formed by the line between the cell start point and the micropipette tip and the line from the cell start to the ending point (Iijima and Devreotes, 2002; Chen et al., 2003), of wild-type (S = 4.38 ± 1.94; CI = 0.96 ± 0.03), *plaA*$^-$ (S = 3.24 ± 1.82; CI = 0.94 ± 0.10), *pi3k1*$^-$/*2*$^-$ (S = 5.60 ± 2.58; CI = 0.92 ± 0.17), and *plaA*$^-$/*pi3k1*$^-$/*2*$^-$ (S = 4.66 ± 2.14; CI = 0.55 ± 0.57) cells were calculated and plotted (n = 15, 37, 37, 49, respectively). Three independent experiments were quantified. (D) Chemotaxis of wild-type and *pi3k1*$^-$/*2*$^-$ cells was observed in the absence or presence of BEL by using phase-contrast microscopy. Cells were treated with BEL (5 μM) for 10 min, then the inhibitor was washed away. Cells were allowed to recover for 5 min before the start of the experiments.

slightly lower, indicating that a fraction of cells were unable to chemotax as well as wild-type cells. In *plaA*$^-$/*pi3k1*$^-$/*2*$^-$ cells, the index was low and the standard deviation was large. About 40% of the cells (i.e., those closest to the pipette) appeared to be more or less chemotactic, while the others moved randomly. Unlike the *plaA*$^-$ cells treated with LY, the speed of random movement of the *plaA*$^-$/*pi3k1*$^-$/*2*$^-$ cells was not significantly reduced compared to those of wild-type cells. These observations suggest that the chemotaxis of the *plaA*$^-$ cells is sensitive to inhibition of PI3K activity.

Next, we asked whether *pi3k1*$^-$/*2*$^-$ cells were more sensitive to inhibition of PLA_2 activity than wild-type cells. Cells were treated with different concentrations of BEL, for 5 min, then washed. In chemotaxis assays, this treatment did not affect the directed movement of wild-type cells, whereas it caused a strong reduction in chemotaxis of the *pi3k1*$^-$/*2*$^-$ cells (Figure 4D; Movies S16–S19).

PLA_2A and PI3K Act in Parallel to Regulate cAMP-Induced Actin Polymerization

As outlined in the Introduction, evidence suggests that the first phase of actin polymerization is regulated by events that act in parallel with the PI3K/PTEN pathway (Chen et al., 2003). To characterize the role of PLA_2A, we carried out biochemical assays of actin polymerization in the *plaA*$^-$ and *plaA*$^-$/*pi3k1*$^-$/*2*$^-$ cells (Figure 5). In the absence of LY, wild-type and *plaA*$^-$ cells displayed similar responses upon cAMP stimulation (Figure 5A). An initial peak (~2-fold above unstimulated cells) occurred at 4–8 s. Since these were highly polarized cells, a defined second peak was not obvious, but the F-actin level remained above baseline for several minutes, as previously described (Chen et al., 2003). Treatment with LY reduced this signal during the prolonged second phase, as expected. The initial peak in the *plaA*$^-$ cells appeared to be slightly decreased as well (Figure 5B). As previously determined, the first peak of actin polymerization was identical in wild-type and *pi3k1*$^-$/*2*$^-$ cells (data not shown). Compared to the *pi3k1*$^-$/*2*$^-$ cells, the first phase of actin polymerization in the *plaA*$^-$/*pi3k1*$^-$/*2*$^-$ cells was clearly decreased (Figure 5C). When treated with LY, a further decrease to 50% of the wild-type level was observed. Therefore, in the *plaA*$^-$ background, the loss of PI3K activity had a detectable affect on the initial peak of actin polymerization. These results suggest that either the PI3K or the PLA_2 pathway can support the initial actin polymerization response, and that inhibition is only observed when both pathways are inhibited.

Purified PLA_2A Converts PC to Arachidonic Acid and Is Inhibited by Calcium

To verify that PLA_2A had PLA_2 activity, we tagged it with a FLAG sequence at the C terminus (PLA_2A-FLAG) and

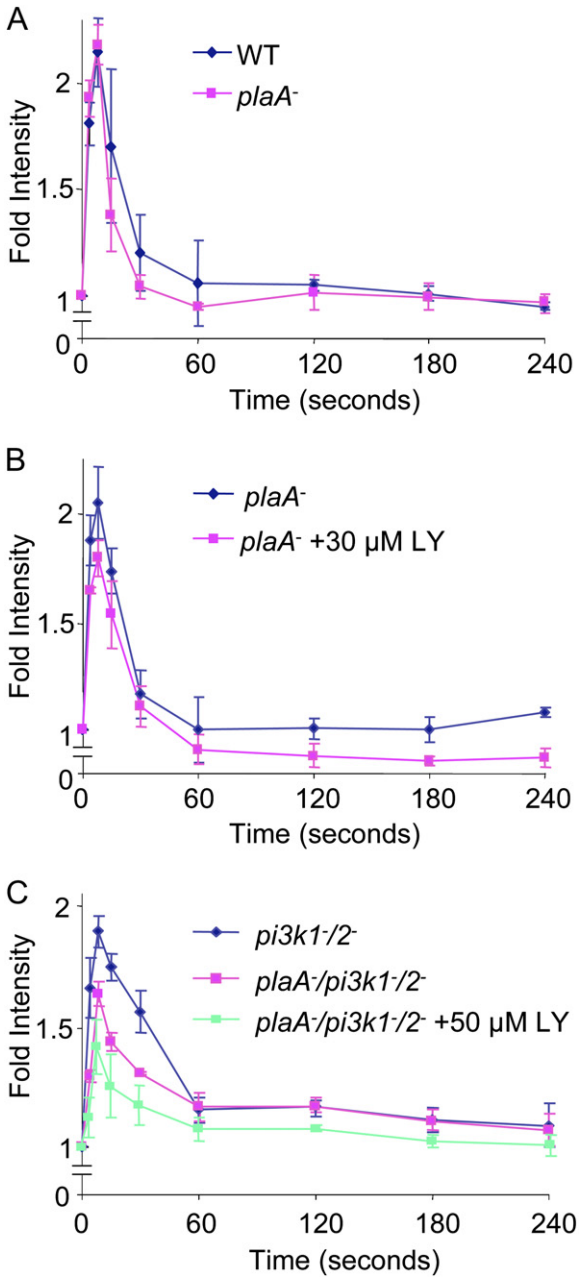

Figure 5. Chemoattractant-Induced Actin Polymerization Assay

(A) Wild-type (diamonds) and *plaA*$^-$ cells (pink squares).

(B) *plaA*$^-$ cells in the absence (blue diamonds) or presence (pink squares) of 30 μM LY.

(C) *pi3k1*$^-$/*2*$^-$ (blue diamonds) and *plaA*$^-$/*pi3k1*$^-$/*2*$^-$ (pink squares) cells in the absence or presence of 50 μM LY (green squares). For each experiment, all values were normalized to the amount of F-actin at time 0, which was taken before the addition of 100 nM cAMP. When indicated, LY was added 10 min before experiments. Experiments were repeated at least three times. Values are means ± standard deviations.

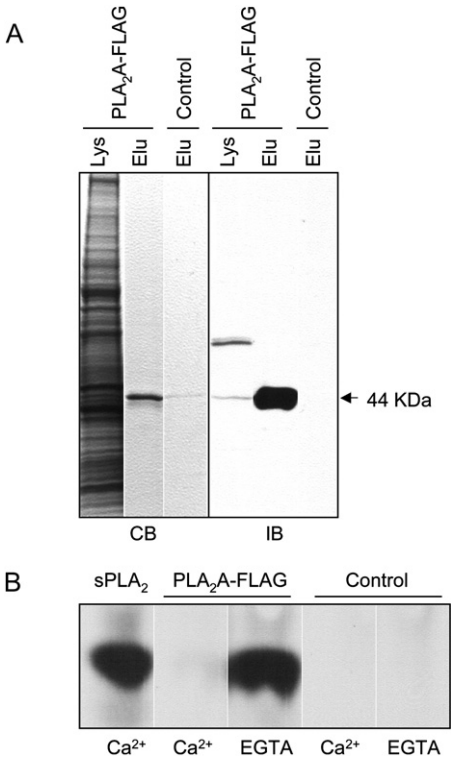

Figure 6. Conversion of Phosphatidylcholine to Arachidonic Acid by Purified PLA_2A-FLAG

(A) PLA_2A-FLAG was affinity purified. A Coomassie blue-stained gel (CB) and immunoblot (IB) show cell lysate and eluted fractions prepared from PLA_2A-FLAG-expressing cells or wild-type cells. The major eluted band from PLA_2A-FLAG-expressing cells is at the predicted size of 44 kD, as indicated by the arrow. A similar-sized, nonspecific band was eluted from wild-type cells but was not recognized by FLAG antibodies. Another nonspecific band (~70 kD) recognized by FLAG antibody is present in the cell lysate, but not in the purified fractions.

(B) Thin-layer chromatography (TLC) of ^{3}H-arachidonic acid (^{3}H-AA) released from ^{3}H-phosphatidylcholine (^{3}H-PC) by purified PLA_2A-FLAG. The eluted fractions were incubated with ^{3}H-PC in buffer containing 5 mM Ca^{2+} or 5 mM EGTA for 60 min at 25°C. ^{3}H-AA generated by a snake venom $sPLA_2$ (Sigma P7778) was used as positive control. The eluted fraction from wild-type cells was used as negative control. The experiment was repeated at least five times.

purified the protein from *D. discoideum* cells. In the under-buffer assay, PLA_2A-FLAG, like PLA_2A-GFP, was able to rescue the LY-sensitive phenotype of the *plaA*$^-$ cells (data not shown). We purified the recombinant protein by ion-exchange chromatography followed by affinity chromatography (Figure 6A). The protein was expressed at low levels, but the recovery from the purification was over 50%, and we were able to obtain a sufficient amount of purified protein (typically 10 μg/10^9 cells) for in vitro PLA_2 assays. The purified PLA_2A-FLAG was able to produce arachidonic acid in buffer containing EGTA or EDTA when phophatidylcholine (PC) was used as substrate (Figure 6B). The specific activity of the purified PLA_2A was similar to that of PLA_2 from snake venom (2.7×10^{-11} μmol/μg·min versus 7.3×10^{-11} μmol/μg·min, respectively) at a substrate concentration of 10^{-9} M. The PLA_2 activities of previously reported $iPLA_2$s, with the exception of the patatin-related PLA_2s, are not activated by Ca^{2+}; thus, they have been called Ca^{2+}-independent

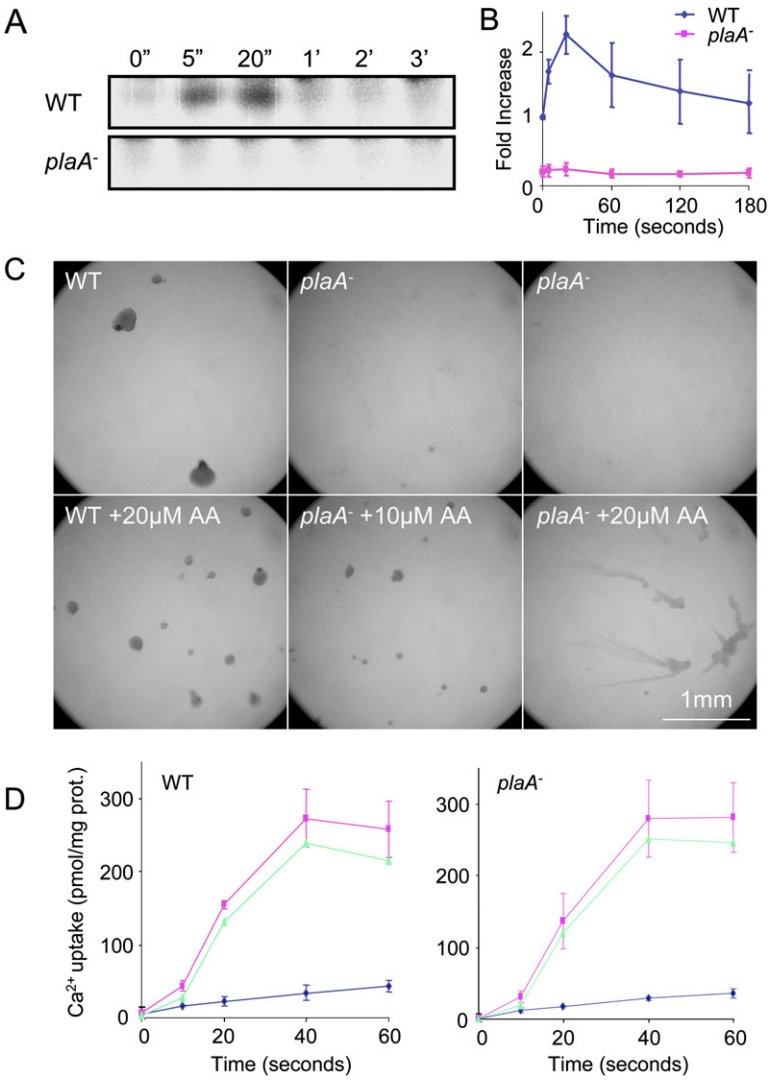

Figure 7. Putative Biochemical Defect in *plaA*⁻ Cells

(A) Thin-layer chromatography (TLC) of a ^{3}H-arachidonic acid (^{3}H-AA)-labeled lipid profile. Wild-type and *pla$_2$a*⁻ cells were labeled with ^{3}H-AA for 2 hr, washed, and subjected to cAMP stimulation. Three major groups of lipids were found on the TLC plate.

(B) Quantification of the spot during cAMP stimulation. Data were averaged from seven independent experiments. Error bars indicate means and standard deviations.

(C) Wild-type and *plaA*⁻ cells in an under-buffer aggregation assay in the presence of 10 μM LY. As indicated, 10 μM or 20 μM AA was added.

(D) Time course of cAMP-induced Ca^{2+} influx of wild-type and *plaA*⁻ cells. Values are shown for resting (blue diamonds), cAMP-stimulated (pink squares), and cAMP-induced (green triangles) Ca^{2+} uptake into cells. Data were calculated from three independent experiments.

PLA$_2$s (Ackermann et al., 1994; Chakraborti, 2002; Holk et al., 2002). The PLA$_2$ activity of PLA$_2$A-FLAG did not require Ca^{2+}; in fact, it was inhibited in buffer containing very high (5 mM) levels of $CaCl_2$ (Figure 6B). The purified enzyme bearing the active site S61A mutation displayed no activity under any condition, consistent with its inability to rescue the mutant phenotype. We did not assess whether the enzyme displayed activity against triacylglycerol and thus have not ruled out that it can also function as a triacylglycerol lipase.

Altered Arachidonic Acid Metabolism Underlies the Phenotype of Cells Deficient in PLA$_2$A

To further elucidate the defect in the *plaA*⁻ cells, we examined the profiles of lipids visualized by steady-state labeling with ^{3}H-arachidonic acid. After several hours of incubation, cells were washed and stimulated with chemoattractant, and lipids were extracted for thin-layer chromatography analyses (Figure 7A). Both wild-type and *plaA*⁻ cells displayed similar levels of the major labeled components, presumably phosphatidylcholine, phosphotidylethanolamine, and other lipids. The wild-type cells also displayed a prominently ^{3}H-labeled component that cochromatographed near arachidonic acid on thin-layer chromatography. Chemoattractants induced a rapid 2- to 3-fold increase in the level of this ^{3}H-arachidonic acid derivative, which peaked within 30 s and declined more slowly (Figure 7B). The levels of the ^{3}H-arachidonic acid derivative were reduced by more than 70% in the *plaA*⁻ cells, and chemoattractant-induced increases were undetectable. The retention factor (Rf) of the compound was somewhat variable (Rf = 0.815–0.903), usually running with or slightly faster than arachidonic acid (Rf = 0.83 ± 0.02). We excised the spot and chromatographed it in an independent TLC system. In the second system, the compound cochromatographed with arachidonic acid (Rf = 0.31) and was widely separated from the methyl ester (Rf = 0.59). These observations suggest that this component is ^{3}H-arachidonic acid or a very closely related, perhaps unstable, derivative.

We wondered whether the severely depressed levels of the arachidonic acid derivative in the *plaA*$^-$ cells might underlie the defect in chemotaxis. We tested whether exogenous arachidonic acid might mitigate the defect. Similar to what was seen in previous under-buffer assays, wild-type cells were unaffected by 10 μM LY, whereas *plaA*$^-$ cells were unable to aggregate in the presence of this inhibitor. However, addition of 5–20 μM arachidonic acid partially reversed this defect (Figure 7C). Exogenous arachidonic acid induced the *plaA*$^-$ cells to aggregate at concentrations of LY as high as 40 μM, whereas wild-type cells were able to aggregate at concentrations up to 80 μM. Oleic acid did not have the same effect as arachidonic acid. These results suggest that the observed deficit in the arachidonic acid derivative may underlie the chemotactic defect.

Published reports have implicated arachidonic acid as a regulator of calcium influx in a variety of cells, including *D. discoideum*. However, as shown in Figure 7D, basal and chemoattractant-mediated Ca^{2+} influxes were very similar in wild-type and *plaA*$^-$ cells. These observations suggest that the effects of PLA_2A on chemotaxis and arachidonic acid metabolism are unlikely to be mediated through effects of calcium uptake.

DISCUSSION

We have found that pathways involving PLA_2 and PI3K act in parallel to mediate chemotaxis. Neither loss of PLA_2A nor significant lowering of $PI(3,4,5)P_3$ levels has a dramatic effect on chemotaxis. However, simultaneous inhibition of both activities caused a severe defect in chemotaxis. The absence of PLA_2A does not affect $PI(3,4,5)P_3$ production, indicating that PLA_2A is not upstream of PI3K. Thus, PLA_2A and its metabolites appear to be part of a parallel pathway regulating chemotaxis and actin polymerization.

These signaling events are involved in the first and second phases of actin polymerization (Chen et al., 2003). The second phase is absent in *pi3k1*$^-$/*2*$^-$ cells, indicating that $PI(3,4,5)P_3$ is the important regulator for the second phase. Our current results suggest that PI3Ks and PLA_2A are equally important for the rapid initial peak; when both are inhibited, the response is significantly decreased. A portion of the initial phase remains even in LY-treated *plaA*$^-$/*pi3k1*$^-$/*2*$^-$ cells, possibly due to our inability to completely remove PLA_2 or PI3K activity, as there are multiple homologs of each enzyme. Alternatively, yet another pathway is involved. It is likely that the decrease in the first peak of actin polymerization contributes to the defect in chemotaxis (Figure 4D). However, other events that play a role in chemotaxis may also be defective.

PLA_2A is more likely to function in the response to chemoattractant gradient than motility per se. The *plaA*$^-$/*pi3k1*$^-$/*2*$^-$ cells exhibit poor chemotaxis but actively form pseudopods and move randomly at the same speed as wild-type cells. LY treatment is apparently not equivalent to deletion of PI3Ks. When first added, LY causes cells to round up; however, when exposed to a chemoattractant gradient, wild-type cells recover within a few minutes, start to extend pseudopods, and move toward the micropipette. Similar results have been observed in several laboratories (P.J.M. Van Haastert and R.A. Firtel, personal communication). However, when treated with LY, the *plaA*$^-$ cells remain rounded for at least 20 min and do not respond to the applied gradient. With longer recovery times, these treated cells regain random motility. Thus, PLA_2A may also be involved in the mechanism by which the applied gradient overcomes the LY-induced motility block.

There are parallel pathways in the chemotaxis network, and each pathway contains multiple redundant components. First, the network is branched after G protein, with PI3K/PTEN regulating one branch and PLA_2A being involved in the other. Inhibition of either pathway does not block chemotaxis since the remaining pathway can compensate. Second, within each pathway, there are several isoforms of the key enzymes. It has been shown that more than two PI3Ks control $PI(3,4,5)P_3$ levels, and we expect that several PLA_2s might also act in concert. These layers of redundancy may, in part, explain why chemotaxis has been difficult to inhibit specifically.

We have identified a specific PLA_2 homolog and have shown that its ability to generate an arachidonic acid derivative may be required for chemotaxis. The observed rapid, chemoattractant-induced increase in the levels of the ^{3}H-arachidonic acid derivative does not occur in *plaA*$^-$ cells. The transient burst of this compound in response to uniform stimulation of wild-type cells is identical to the behavior of other molecules, such as $PI(3,4,5)P_3$, that have been implicated in directional sensing and have been found at the cell's leading edge when the stimulus is applied as a gradient. Further studies are required to precisely identify the derivative and determine its location in chemotaxing cells. The strong defect in arachidonic acid metabolism is remarkable considering that there are 14 patatin-like PLA_2s in *D. discoideum*. It is possible that PLA_2A targets a specific phospholipid substrate or phospholipids within a specific compartment. The chemoattractant regulation and specific defect in the mutant indicates that PLA_2A plays more than a "housekeeping" role.

Exogenous arachidonic acid (and not oleic acid) can compensate for the loss of PLA_2A, indicating a specific requirement for an achidonic acid metabolite. Uniform application of arachidonic acid can decrease sensitivity to PI3K inhibition, even though it would not be expected to produce an internal gradient within the cell. Perhaps arachidonic acid enhances the activity of another signaling molecule that is asymmetrically localized. While a localized increase in arachidonic acid normally synergizes with the other component to produce an optimal response, uniform levels may be able to partially function by enhancing the other activity. Alternatively, other products of PLA_2A, such as a lyso-phospholipid, may also be required for full rescue.

We have as yet no insight as to the target of the PLA_2A metabolites. Studies in mammalian cells have suggested that arachidonic acid may regulate store-operated

calcium channels (Smani et al., 2004). Furthermore, previous studies in *D. discoideum* have suggested that products of PLA_2, arachidonic acid, and platelet activation factor (PAF, synthesized from lyso-phosphatidylcholine) affect chemoattractant-induced Ca^{2+} influx and can trigger Ca^{2+} influx directly (Bussoloni et al., 1991; Sordano et al., 1993; Schaloske et al., 1995; Schaloske and Malchow, 1997). Ca^{2+} influx stimulated by cAMP was blocked with PLA_2 inhibitors and was then bypassed by arachidonic acid and related components (Schaloske et al., 1998). However, we found that specific genetic disruption of PLA_2A that leads to a chemotaxis defect and altered arachidonic acid metabolism does not affect Ca^{2+} uptake. Other studies have also not supported a role for Ca^{2+} in chemotaxis. Cells lacking a putative IP3 receptor homolog (*iplA*$^-$), in which Ca^{2+} influx in response to cAMP is abolished, displayed no chemotaxis defect (Traynor et al., 2000). Consistently, we found that *iplA*$^-$ cells were not more sensitive than wild-type cells to inhibitors of PI3K (data not shown).

Our studies provide genetic evidence that PLA_2 and PI3K activities act in parallel. Similar mechanisms might exist in other organisms. Arachidonic acid-releasing phospholipases are enriched in nerve growth cones and have been implicated in neurite outgrowth (Hornfelt, et al., 1999). Studies of chemotaxis in monocytes revealed requirements for $cPLA_2$ and $iPLA_2$ with distinct characteristics (Carnevale and Cathcart, 2001). Inhibition of $cPLA_2$ could be overcome by exogenous arachidonic acid, as shown in our studies. It is likely that such parallel pathways are conserved and that PLA_2 is a central regulator of chemotaxis along with PI3K/PTEN. Consideration of the parallel roles of these pathways may help to resolve some of the conflicting results observed by inhibition of PI3K in other systems.

EXPERIMENTAL PROCEDURES

Cell Culture, Development, and Mutagenesis

D. discoideum cells were cultured in HL5 medium and allowed to differentiate for 5 hr, unless otherwise indicated, in development buffer (DB) as previous described (Parent et al., 1998). To isolate mutants sensitive or resistant to LY294002, wild-type cells were mutagenized, and genes were identified by using the restriction enzyme-mediated integration (REMI) method (Adachi et al., 1994; Van Es et al., 2001).

Live Cell Imaging and Quantification

Fluorescent images of living cells expressing GFP fusion proteins and chemotactic movements of cells toward cAMP-containing micropipettes were gathered as previously described (Parent et al., 1998). IP lab, Image J, and the Matlab imaging tool box (Mathworks) were used to collect and process data (Chen et al., 2003).

Cellular Responses to Chemoattractant Stimulation

PH domain translocation, actin polymerization, and calcium influx assays were performed as previously described (Parent et al., 1998; Iijima and Devreotes., 2002; Zigmond et al., 1997; Milne and Coukell, 1991).

Protein Purification and Phospholipase A_2 Assays

Wild-type cells expressing PLA_2A-FLAG were cultured to a density of 3–8 × 10^6 cells/ml. Typically, 500 ml cells were collected and starved at 2 × 10^7 cells/ml for 2 hr, collected, and filter lysed in 50 mM HEPES (pH 7.5) at a density of 1 × 10^8 cells/ml (Parent and Devreotes, 1999). Cell lysates were subjected to two rounds of centrifugation at 15 Krpm for 20 min, and the supernatant was centrifuged at 55 Krpm for 20 min. The final supernatant was loaded on an ion-exchange column (Q fast flow, Amersham). The Q column was washed with 0.1 M NaCl with 50 mM HEPES (pH 7.5) and was then eluted with 0.5 M NaCl with 50 mM HEPES (pH 7.5). The eluted fraction (3–4 ml) was incubated with 200 μl FLAG-agarose (Sigma) for 2–3 hr at 4°C. Agarose beads were collected, washed, and incubated at 4°C for 10 min with 400 μl 200 ng/μl FLAG-peptide (Sigma) in 100 mM HEPES, 0.1% Triton X-100. After centrifugation, the supernatant was collected and subjected to further analysis. In some experiments, 10 mM sodium phosphate buffer (pH 7.0) was used instead of 50 mM HEPES. PLA_2 assays were performed as previously described, with minor modifications (Ackermann et al., 1994). Extracted products were separated on a Silica gel 60 TLC plate (EMD chemicals) in chloroform:methanol:acetic acid:-water (75:20:2:1, v/v/v/v). Then, the TLC plate was sprayed with ^{3}H enhancer (PE) and exposed to HyBlot film (Denville) at −80°C for 2 days.

^{3}H-Arachidonic Acid Labeling Assay

Cells were starved for 3 hr in DB and were labeled with ^{3}H-arachidonic acid for another 2 hr. Labeled cells were resuspended at 3 × 10^7 cells/ml in DB and shaken at 200 rpm at room temperature. At various time points after adding 500 nM cAMP, 300 μl cells were collected into 1 ml chloroform:methanol:acetic acid (2:4:1, v/v/v) to stop the stimulation. Lipids were extracted and subjected to TLC analyses as described in the previous section.

Supplemental Data

Supplemental Data include 2 figures and 19 movies and are available at http://www.developmentalcell.com/cgi/content/full/12/4/603/DC1/.

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